

Detecting Deepfake Audio Using Spectrogram-Based Machine Learning Approaches

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Abstract

Spectrogram visual representations offer rich time-frequency information for deepfake analysis. This paper conducts a rigorous empirical investigation comparing Log-Mel, MFCC, Constant-Q Transform (CQT), and Gammatone Spectrograms paired with Convolutional Neural Networks (ResNet-50, EfficientNet-B0) to classify authentic vs deepfake speech.

Key Methodology & Architecture

Time-frequency spectrogram transformation (Log-Mel, CQT, MFCC), ResNet-50, EfficientNet, Transfer Learning, Grad-CAM visual explainability.

Datasets & Benchmark Environments

FoR (Fake or Real) Dataset, ASVspoof 2021 LA.

Performance Metrics & Graph Axis Parameters

Reported Results: CQT + EfficientNet Accuracy = 98.42%, F1-Score = 0.984, Precision = 0.986, AUC = 0.995.

Graph Parameter Mapping: X-axis: Feature Representation (Log-Mel, MFCC, CQT, Gammatone); Y-axis: Classification Accuracy (% from 80% to 100%).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Visual explainability via Grad-CAM highlighting synthetic high-frequency noise.	Spectrogram generation requires preprocessing overhead.

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